Pietro Carlo Boldini

Postdoctoral Researcher · Delft University of Technology

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Education

10/2020 – 01/2026 **PhD, Mechanical Engineering**, Delft University of Technology (NL)
Process & Energy Department, Faculty of Mechanical Engineering.
Dissertation: *Boundary-Layer Transition with Fluids at Supercritical Pressure.* Advisor: Prof. Dr. R. Pecnik.

04/2016 – 12/2018 **M.Sc., Aerospace Engineering**, University of Stuttgart (DE)
GPA: 4.0/4.0 (German grade 1.2, with distinction, top 3%).
Master's Thesis (Laboratory for Advanced Simulation of Turbulence, Tsinghua University, Beijing): *Effects of Thermal Non-Equilibrium on Hypersonic Boundary-Layer Transition.* Supervisors: Prof. Dr. S. Fu, Dr.-Ing. M. J. Kloker.

10/2012 – 04/2016 **B.Sc., Aerospace Engineering**, University of Stuttgart (DE)
GPA: 3.7/4.0 (German grade 1.5).
Bachelor's Thesis (Institute of Space Systems): *Optimum Design of the Intake for an Atmosphere-Breathing Electric Propulsion System.* Supervisors: Dr.-Ing. F. Romano, Dr.-Ing. T. Binder, Prof. Dr.-Ing. G. Herdrich.

Publications

Selected Journal Articles

[J5] P. C. Boldini, R. Hirai, B. Bugeat, and R. Pecnik. *First-Order Buoyancy Correction of Modal Instabilities in Stratified Boundary Layers.* **Journal of Fluid Mechanics**, 1036, A39, 2026 (DOI).

[J4] P. C. Boldini, B. Bugeat, J. W. R. Peeters, M. Kloker, and R. Pecnik. *Direct Numerical Simulation of Complete Transition to Turbulence with a Fluid at Supercritical Pressure.* **Journal of Fluid Mechanics**, 1025, A59, 2025 (DOI).

[J3] P. C. Boldini, R. Hirai, P. Costa, J. W. R. Peeters, and R. Pecnik. *CUBENS: A GPU-Accelerated High-Order Solver for Wall-Bounded Flows with Non-Ideal Fluids.* **Computer Physics Communications**, 309, 109507, 2025 (DOI).

[J2] P. C. Boldini, B. Bugeat, J. W. R. Peeters, M. Kloker, and R. Pecnik. *Transient Growth in Diabatic Boundary Layers with Fluids at Supercritical Pressure.* **Physical Review Fluids**, 9, 083901, 2024 (DOI).

[J1] B. Bugeat, P. C. Boldini, A. M. Hasan, and R. Pecnik. *Instability in Strongly Stratified Plane Couette Flow with Application to Supercritical Fluids.* **Journal of Fluid Mechanics**, 984, A31, 2024 (DOI).

See full list: Google Scholar, ResearchGate

Research Experience

09/2025 – present **Postdoctoral Researcher**, Aerodynamics Group, Faculty of Aerospace Engineering, Delft University of Technology (NL)
Project: *Flow Control through Metamaterials for Transition Delay and Drag/Noise Reduction.* Supervisor: Prof. Dr. M. Kotsonis.

04/2019 – 03/2020 **Research Associate**, Institute of Aerodynamics and Gas Dynamics, University of Stuttgart (DE)
Project: *Cross-flow transition control in a hypersonic boundary layer on conical geometries with thermal non-equilibrium effects.* Supervisor: Dr.-Ing. M. J. Kloker.

2017 – 2019 **Student Research Assistant**, Institute of Aerodynamics and Gas Dynamics, University of Stuttgart (DE)
Project: *DNS of compressible turbulent boundary-layer flows.* Supervisor: Dr.-Ing. C. Wenzel.

2016 – 2017 **Student Research Assistant**, Institute of Space Systems, University of Stuttgart (DE)
Project: *Numerical modeling for Atmosphere-Breathing Electric Propulsion (EU Project DISCOVERER).* Supervisors: Dr.-Ing. F. Romano, Dr.-Ing. T. Binder.

Software

CUBENS **CUBic Equation of state Navier–Stokes solver** – GPU-accelerated (NVIDIA & AMD) high-order DNS solver for wall-bounded flows with non-ideal fluids. Open source: github.com/pcboldini/CUBENS

DeHNSSo-v2 **Delft Harmonic Navier–Stokes Solver** – nonlinear harmonic Navier–Stokes framework for stability problems in boundary layers under development during my postdoctoral work. Extended version of Westerbeek et al. (CPC, 2024, DOI): curvilinear coordinates, discrete adjoint, compressible extension, GPU acceleration. Paper in preparation.

DragHeat-NIF **Drag and Heat Transfer Estimation for Non-Ideal Fluids** – estimates mean profiles, skin friction, and heat transfer in turbulent boundary layers with non-ideal fluids via inner/outer-layer transformations. Open source: github.com/pcboldini/DragAndHeat-NIF

Teaching & Supervision

2023 – 2024 Teaching Assistant, *Advanced Heat Transfer* (ME45001), M.Sc. Mechanical Engineering, TU Delft.

2024 M.Sc. Thesis co-supervisor: F. Ramalho Matias, *Three-Dimensional Linear Stability Analysis of Flat-Plate Boundary Layers with Supercritical Fluids*, TU Delft.

2023 M.Sc. Thesis co-supervisor: R. Gaspar (University of Liege), *Investigation of the Transition to Turbulence on a Flat-Plate Boundary Layer at Supercritical Pressure*, TU Delft.

2023 M.Sc. Thesis co-supervisor: J. Wang (Sorbonne University), *Secondary instabilities in flat-plate boundary-layer flow for supercritical fluids*, TU Delft.

2022 M.Sc. Thesis co-supervisor: P. R. Molahalli, *Secondary Flows in Asymmetrical Contractions. A Numerical Study*, TU Delft.

Selected Talks & Presentations

[T8] P. C. Boldini, R. Hirai, P. Costa, J. W. R. Peeters, and R. Pecnik. *CUBENS: A GPU-accelerated high-order solver for wall-bounded flows with non-ideal fluids.* Computer Physics Communications Seminar Series, 2025. **(Invited)**

[T7] P. C. Boldini, B. Bugeat, J. W. R. Peeters, M. Kloker, and R. Pecnik. *Widom-line effect on the boundary-layer transition with a highly non-ideal fluid.* 16th ERCOFTAC SIG33 Workshop, Cagliari, Italy, 2025.

[T6] P. C. Boldini, B. Bugeat, J. W. R. Peeters, M. Kloker, and R. Pecnik. *DNS of K-type transition in a flat-plate boundary layer with supercritical fluids.* IUTAM Symposium on Laminar-Turbulent Transition, Nagano, Japan, 2024.

[T5] P. C. Boldini, R. Hirai, P. Costa, J. W. R. Peeters, and R. Pecnik. *GPU-accelerated DNS of diabatic transitional boundary layers at supercritical pressures.* DLES 14, Erlangen, Germany, 2024.

[T4] R. Hirai, P. C. Boldini, B. Bugeat, R. Pecnik. *DNS of H-type transition in stratified horizontal boundary layers.* 77th APS Division of Fluid Dynamics, Salt Lake City, USA, 2024. (presented by R. Hirai)

[T3] P. C. Boldini, B. Bugeat, P. Costa, J. W. R. Peeters, and R. Pecnik. *DNS of K- and H-type transitions in a flat-plate boundary layer with supercritical fluids.* 76th APS Division of Fluid Dynamics, Washington D.C., USA, 2023.

[T2] P. C. Boldini, B. Bugeat, P. Costa, J. W. R. Peeters, and R. Pecnik. *DNS of H-type transition in a flat-plate boundary layer with supercritical fluids.* 21st STAB-Workshop, DLR Göttingen, Germany, 2023.

[T1] P. C. Boldini, B. Bugeat, J. W. R. Peeters, and R. Pecnik. *Transient growth in flat-plate boundary layers with a highly non-ideal fluid.* 15th ERCOFTAC SIG33 Workshop, Alghero, Italy, 2023.

Computational Skills

Simulation	DNS, linear/nonlinear stability analysis, OpenFOAM
HPC & GPU	CUDA (NVIDIA & AMD); OpenACC; MPI; large-scale HPC clusters
Programming	Fortran, MATLAB, Python, C++, Bash/Shell
Workflow	Git/GitHub; LaTeX; Linux/macOS

Awards & Funding

2024	Co-author, NWO Computing Time Grant (PI: Prof. Dr. R. Pecnik), *Direct Numerical Simulations of Turbulent Heat Transfer with Non-Ideal Gases and Surface Roughness.* 30M GPU hours & 10M CPU hours on the Dutch National supercomputer Snellius (SURF NL).
2018	DAAD scholarship for Master's Thesis research at Tsinghua University, Beijing.
2014	Top Performer Award in Astronautics I, top 5% of 111 participants, B.Sc. Aerospace Engineering, University of Stuttgart.

Peer Review

Referee	Journal of Fluid Mechanics (3); Physical Review Fluids (1); Physics of Fluids (1); Global Power and Propulsion Society Technical Conference (1)

Delft, June 4, 2026